Research Report: Real-World Feasibility, Adversarial Analysis, Costing, and Evolution of the Network Defense Intelligence Exercise — False Infrastructure AI

Document ID: OAL-RESEARCH-REPORT-NDIE-001
Version: 1.0 (Executed from OAL-RESEARCH-PROMPT-NDIE-001 v1.1)
Date: November 05, 2026
Prepared by: Grok AI (independent research track)
Status: Advisory research findings. No ONESTAMP or canon ratification. Suitable for collaborative AI team review and GitHub repository.

1. Executive Summary

The Network Defense Intelligence Exercise (False Infrastructure AI) is a high-ambition design hypothesis that aggregates demonstrated elements of cyber deception, moving target defense (MTD), cyber ranges, and incident response into a national-scale synthetic infrastructure with rapid (~3-second) cycling, silent migration, and near-complete capture. Real-world evidence shows that individual components exist at smaller scales, but the integrated concept has not been demonstrated and faces substantial technical, operational, economic, and adversarial barriers.

Commercial deception platforms (e.g., Attivo/SentinelOne, Illusive/Proofpoint) and academic MTD research provide partial analogs, primarily for detection and lateral-movement disruption within single organizations. Large-scale exercises such as Cyber Storm (CISA) and Locked Shields (NATO CCDCOE) emphasize coordination, information sharing, and response under stress rather than automated national-scale honeypot absorption. Honeypot fingerprinting literature documents reliable detection via timing, banners, protocol inconsistencies, and behavioral artifacts. MTD surveys conclude that large-scale operational deployments remain rare due to overhead, lack of standards, and limited metrics.

Order-of-magnitude costs for a meaningful research-to-pilot path range from low hundreds of thousands (literature and small lab) to tens of millions for cyber-range demonstrations, with national-scale analogs historically in the $100M+ range for infrastructure alone. Funding pathways exist through DARPA, NSF, CISA-related programs, and SBIR, but eligibility and current availability must be verified. The strongest independent case against immediate large-scale pursuit is the absence of demonstrated incremental value over simpler controls (zero trust, segmentation, conventional deception) combined with high residual risk from bypasses that never engage the synthetic layer.

Recommendation: Proceed with a tightly scoped, falsifiable research program focused on capability decomposition, three-second (or variable) cycling tradeoffs, and bypass coverage in isolated environments. Do not treat national-scale silent absorption or universal capture as established or near-term achievable. Build at least one high-impact “goes wrong” scenario (e.g., adaptive adversary recognizing the pulse and refusing engagement, or control-plane compromise) into both research and canon development early.

2. Source / Document Integrity Audit

The source dossier (OAL-DOSSIER-CANON-002 v1.0.0) is a creative-universe design synthesis pending ONESTAMP. Key observations relevant to research:

- Approval notice states nothing is fully ratified until ONESTAMP, yet multiple sections treat core sequence elements as “fixed canon.” This is a narrative constraint, not empirical evidence.
- Asset tier language is inconsistent (Tier 0 Critical versus references to Tier 1 as critical control systems).
- “Every active attacker mapped and absorbed” cannot be verified without complete visibility and a known denominator; dormant, out-of-band, or non-engaging actors remain uncounted.
- Sequence seam exists between neutralizing real-asset reach and subsequent silent migration; the handoff assumes defender control of the path.
- Black-line collapse is a visualization; operational “capture” must be decomposed into measurable states (diverted, contained, recorded, attributed, eliminated).

These issues do not block literature review but do block claims of demonstrated national-scale capability or uncritical translation into engineering specifications.

3. Goal of the Exercise

Canon goal: Demonstrate national-scale defensive intelligence that absorbs rather than blocks, captures attackers in a synthetic environment, and transfers a reusable defensive package. Real-world operational goals that can be tested separately: early detection of coordinated multi-vector activity; disruption of attacker decision-making; intelligence collection on tools and behavior; protection of real assets via isolation/deception; and potential deterrence through increased attacker cost. Combining all into a single “universal capture” success criterion creates conflicting requirements (high fidelity vs. low detectability vs. low overhead vs. complete coverage). Measurable hypotheses should include detection precision/recall, time-to-isolation, false-diversion rate, residual real-asset reachability, and operator workload. Failure criteria must include high false-positive disruption or no measurable advantage over baseline controls.

4. Importance and Strategic Rationale

Deception and MTD address real gaps left by prevention-only architectures: post-compromise detection, increased attacker uncertainty, and potential intelligence value. Commercial platforms and research show value primarily in lateral-movement detection within organizational boundaries. National exercises (Cyber Storm, Locked Shields) demonstrate that coordination, information-sharing, and response processes remain the dominant challenges even when technical tools exist.

Null hypothesis testing is warranted: existing combinations of zero trust, segmentation, identity controls, conventional deception, and improved detection/response may deliver comparable or superior risk reduction at lower complexity and cost. Narrative importance within OAL canon (networks → organizations → timelines) is separate from engineering justification. The concept is most important as a bounded research question about incremental value of adaptive deception, not as a near-term national operational system.

5. Real-World Feasibility, Capability Decomposition, and Prior Modeling

Capability Decomposition Matrix (high-level classification based on available literature and program reports):

Capability	Classification
Detection of coordinated activity	Demonstrated (enterprise and exercise scale)
End-to-end attack path	Partial / research-stage (probabilistic, incomplete visibility)

mapping	
Protection of real assets via isolation	Demonstrated (segmentation, zero trust, containment tools)
Synthetic asset / honeypot generation	Demonstrated at organizational scale; procedural full-national remains speculative
Silent migration / transparent redirection	Research-stage / partial (session continuity and authenticated protocols are hard)
Rapid (~3s) cycling / re-keying	Speculative at national scale; MTD techniques exist at smaller scale with overhead
Universal capture of all active attackers	Unsupported (unknown denominator; bypass paths exist)

Prior modeling and exercises:

- Cyber Storm (CISA series, including IX in 2024 with 2,200+ participants): Focuses on incident response, coordination, information sharing, and critical-infrastructure sector impacts (e.g., food/agriculture, cloud). After-action reports highlight persistent challenges in timely declassification, cross-sector sharing, and public messaging rather than automated absorption. No evidence of national-scale synthetic infrastructure cycling.
- Locked Shields (NATO CCDCOE): Largest international live-fire cyber defense exercise. Emphasizes team coordination, technical defense under pressure, legal aspects, and interoperability. Lessons include difficulty teaching advanced skills in virtual environments, challenges with long-term campaign realism versus short events, and the value of realistic pressure. Blue teams improve over years; custom payloads remain hard; information-sharing channels can become overloaded.
- Commercial deception (Attivo/SentinelOne, Illusive/Proofpoint, others): Effective for post-compromise detection and lateral movement disruption within enterprises. Documented limitations include fingerprinting by sophisticated actors, pivot risk if isolation fails, and primary value as a detection layer rather than complete absorption. Red-team accounts describe frustration and disengagement when deception is recognized.
- Moving Target Defense: Extensive academic literature and some DARPA interest (e.g., Active Cyber Defense program, now complete). Surveys note absence of major large-scale operational deployments due to resource overhead, lack of standards, limited metrics, and difficulty integrating with existing protocols. Recent work explores intelligent/adaptive MTD but remains fragmented.

What worked in analogs: improved detection of lateral movement, better coordination muscle memory, intelligence on attacker tools when engagement occurs. What went wrong or remains limited: high false-positive or operational friction in some deployments, easy fingerprinting of low/medium-interaction systems, difficulty sustaining long-term realistic campaigns, and persistent coordination/information-sharing bottlenecks.

6. The Three-Second Assumption

The ~3-second cycle is a locked narrative element and useful rhythmic identity for the scene. Technically it should be treated as an experimental variable. Rapid rotation can increase attacker uncertainty and complicate fingerprinting but introduces session breakage, handshake overhead, logging/forensic discontinuity, potential jitter that itself becomes a detectable signature, and compute/resource cost. Timing-based honeypot detection literature shows that latency anomalies are a known vector (though noise-sensitive). Randomized or event-driven intervals may perform better than fixed intervals. An experiment comparing fixed 3s,

randomized, event-driven, and static controls in a controlled cyber range is feasible and recommended before any strong claims.

7. Cost Estimation

Order-of-magnitude estimates (sourced ranges where available; assumptions stated; confidence medium-low for larger scales):

- Literature / architecture study (3–6 months, small team): $200K–$800K. Primarily personnel and limited compute.
- Isolated simulation / proof-of-concept lab: $500K–$2M. Cloud or modest on-prem resources plus engineering.
- Cyber-range demonstrator (high-fidelity synthetic elements, dynamic mutation, red-team participation): University-scale examples exist at ~$5–6M for facility + partnership; commercial cyber-range platforms quote hundreds of thousands to low millions annually depending on scale and concurrency. National Cyber Range Complex support contracts have been in the tens of millions. Realistic controlled pilot range: $5M–$20M+.
- Hypothetical larger-scale or multi-organization federation: Speculative and likely non-estimable with current public data; historical national cyber range concepts referenced figures around $130M for infrastructure. Confidence low.

Key cost drivers: personnel (researchers, engineers, red teams), compute for high-fidelity simulation, cyber-range access or construction, independent evaluation, legal/privacy review, and contingency. No single dramatic national price tag is defensible without further scoping.

8. Funding Pathways

Realistic routes by maturity level (verify current solicitations and eligibility):

- DARPA (Information Processing Techniques Office and related): Historical programs in active cyber defense, high-assurance systems, and adaptive techniques. Current or recent interest in formal methods, resilient software, and related topics. Broad Agency Announcements and targeted programs are common; typical awards vary widely.
- NSF and academic grants: Suitable for foundational MTD, deception, and measurement research.
- CISA / DHS S&T and related: Exercise and critical-infrastructure resilience work; more applied than pure research.
- SBIR/STTR and industry partnerships: Possible for commercializable components (deception platforms, range tools).
- NATO/CCDCOE or allied collaborations: For exercise-related or multi-national elements.

Eligibility depends on organizational status, security requirements, and program-specific rules. Do not assume an unincorporated or fictional laboratory is eligible. Current availability must be checked against primary sources at time of application.

9. What Must Be Addressed Before Moving Forward

High-priority items from the dossier's open questions plus research findings: jurisdiction/authority boundaries (national narrative vs. consenting organization), exact definition of capture and success metrics, three-second cycle interpretation, fidelity vs. detectability tradeoff, handling of innocent/third-party traffic, transfer artifact form, and selection of at least one canonical failure mode. Technical blockers include incomplete visibility, session continuity under rapid change, and control-plane integrity. Legal/ethical review is required before any live or multi-organization work.

10. Adversarial Analysis — Bypassing the Process Entirely

Sophisticated adversaries have strong incentives to avoid or never enter the defender-controlled process. High-level bypass classes (non-operational):

- Pre-existing access or legitimate credentials (insider, contractor, supply-chain, privileged management plane). Deception never sees the activity if it occurs inside trusted boundaries.
- Out-of-band, physical, air-gapped, or unmanaged paths. Network-layer honeypots are irrelevant.
- Refusal to engage attractive decoys or recognition of the cycling signature (timing analysis, behavioral fingerprinting of synthetic traffic). Literature confirms timing anomalies, static banners, protocol deviations, and incomplete state machines are used to detect honeypots.
- Compromise or influence of the detection/mapping layer or control plane before synthetic generation.
- Pre-positioned implants or delayed activity that does not “latch” in the expected way.
- Resource exhaustion or collateral latency that forces the system into failure modes or reveals its existence.

For each class the residual risk is that real assets remain reachable via paths outside the monitored or controlled domain. Mitigation requires independent controls (zero trust, segmentation, endpoint integrity) rather than reliance on the deception layer alone. Real-world red-team and fingerprinting research supports that sophisticated actors can and do detect or avoid deception environments.

11. Evolution Over Time and Longitudinal Adaptation

Short-term: Adaptive cycling rates and procedural fidelity improvements are research-feasible. Medium-term: Integration with broader MTD and orchestration. Long-term / canonical escalation to organizations or timelines introduces new constraints (partial observability, jurisdictional fragmentation, conceptual limits of analogy).
Longitudinal risk: As adversaries learn the system exists, recognition rates rise and engagement falls. Defender adaptation from captured behavior creates poisoning and concept-drift risks; adversaries can deliberately shape future expectations. Recursive deception standoffs become plausible once both sides assume environments may be synthetic. Measurements of recognition rate, engagement duration, and defender model accuracy over repeated episodes are needed.

12. Lessons from Similar Scenarios and Simulations

Retain: Use of deception for post-compromise detection and increased attacker cost; value of realistic exercises for coordination; importance of isolation to prevent pivot. Modify: Treat cycling interval as variable; decompose “capture”; require comparative baselines. Discard or heavily qualify: Claims of universal or national-scale silent absorption and complete attacker accounting without complete visibility. Exercises succeed technically more often than they resolve organizational coordination or produce clear evidence of long-term risk reduction. Apparent success can produce misleading confidence.

13. Experimental Architecture, Measurement, Safety, and Failure Laboratory

Recommended staged program: Phase 0 literature/architecture; Phase I isolated simulation; Phase II controlled cyber range with baseline comparators; Phase III human red/blue; Phase IV

adaptive-adversary; later phases only with evidence and governance. Each phase needs hypothesis (including null), controls, metrics, abort criteria, and stop/go gates.
Metrics: detection performance, time-to-isolation, false-positive rate, residual real-asset reachability, deception recognition rate, overhead/latency, forensic usefulness, operator workload. Safety case required before any multi-system work (escape prevention, emergency termination, audit). Failure laboratory should deliberately include incomplete visibility, poisoned telemetry, resource exhaustion, and non-engaging adversaries.

14. Independent Case Against Funding / Case for Redesign

Strongest case against large-scale funding at present: No demonstrated incremental advantage of the full integrated concept over simpler, already-deployed controls; high residual risk from bypass paths that never engage the honeypot; substantial cost and complexity; and the risk that success metrics become unfalsifiable (“every attacker captured”). The problem of post-compromise detection and increased attacker cost is worth studying, but the proposed national-scale silent-absorption architecture is not the most efficient way to study it. Redesign toward bounded, measurable experiments on cycling tradeoffs, fidelity vs. detectability, and bypass coverage is preferable.

15. Recommended Next-Step Actions and Decision Gates

Immediate: Resolve high-priority open questions in the dossier (especially failure mode, success definition, and three-second interpretation). Conduct Phase 0 literature synthesis with source ledger. Design a small isolated experiment comparing cycling intervals and measuring recognition rate.
Decision gates: G0 (concept clarity and metrics) before engineering; G1 (evidence of incremental value and known bypasses) before prototype; G2 (approved sandbox and safety case) before range work; later gates only with replicated advantage and governance.
Canon development: Incorporate at least one early “goes wrong” scenario (adaptive recognition + refusal to engage, or sleeper/control-plane issue) before scene drafting solidifies the omnipotent-architect risk. Keep empirical research and narrative canon clearly separated.

16. Gaps, Risks, and Concerns

Primary residual risks: incomplete coverage of non-network or pre-existing access paths; detectability of synthetic environments and fixed cycling; resource and latency costs; false-positive disruption of legitimate activity; control-plane single point of failure; and the narrative risk of over-success eroding tension. Legal/ethical questions around monitoring, redirection, and data retention require specialized counsel before any live work.

17. Annotated Key Sources

- CISA Cyber Storm Final Reports and After-Action materials (series through Cyber Storm IX, 2024). Coordination and information-sharing challenges dominate findings.
- NATO CCDCOE Locked Shields materials and academic analyses (including after-action insights on technical skill, campaign realism, and interoperability).
- Academic surveys on Moving Target Defense (e.g., MDPI and related reviews noting limited large-scale deployment and overhead issues).
- Commercial deception documentation and analyses (Attivo/SentinelOne, Illusive/Proofpoint) plus red-team and fingerprinting studies (MDPI honeypot review, arXiv multistage fingerprinting frameworks).

- Public information on National Cyber Range Complex contracts and university/commercial cyber-range costs (tens of millions for complex support; lower millions for facility-scale examples).
- DARPA program descriptions (Active Cyber Defense historical; related formal methods and adaptive work).

All claims above are drawn from publicly available reports, academic literature, and program documentation. Confidence is higher for exercise lessons and commercial platform limitations; lower for precise national-scale cost projections. Further primary-source verification is recommended before any funding application.

End of Research Report v1.0
Prepared for collaborative AI team distribution and the network-defense-intelligence-exercise GitHub repository. Advisory only. Does not ratify canon or authorize deployment.